**Patient ID:** TCGA-A5-A0G2
**Age:** 57 at diagnosis
**Sex:** Female

**Clinical Stage:** IIIB
**Data Type:** Cross-sectional
**Researcher:** Tori Sauve

## Clinical Presentation

TCGA-A5-A0G2 was diagnosed with stage IIIB endometrial carcinoma (Serous endometrial adenocarcinoma). This patient has cross-sectional data, resulting from a single biopsy from the endometrial site, with no longitudinal data available. As a result, it is possible to determine key mutations driving the cancer and identify potential pathways of cancer progression by comparing to theoretical frameworks for progression of endometrial cancer. This is further complicated by the unique mutational load of the patient, with the biopsy revealing 22,936 mutations. It is impossible to determine the exact mutational progression of this patient's tumor even with longitudinal data, which ordinarily show the order in which mutations appeared. The result is common for POLE ultra-mutated cancers as mutations accumulate rapidly and in large numbers[1].

## Molecular Interpretation

Cohort comprises endometrial carcinoma patients with ten thousand or more mutations, resulting in 10 patients for comparison.

- Biopsy type: single (cross-sectional) samples as opposed multiple samples taken over the patient's cancer progression (longitudinal)
- Patient's Mutational load: 22,936 somatic variants
- Mutation Frequency Analysis in cohort: All patients exhibit POLE mutations with V411L, P286R, and P916L occurring with the most frequency. Other POLE mutations in the cohort appear once (See Figure 1, Appendix I)
- Patient is POSITIVE for POLE V411L in addition to S2084L, P696S, M562T, and R114*
- Impression: The patient exhibits a mutational load outside of that predicted by the Bokhman Dualistic Model[2]. This suggests that the patient falls into the POLE ultra-mutator subtype. However, unlike colorectal carcinoma, for which the Vogelstein-Fearon model establishes a canonical mutational sequence, POLE ultra-mutated endometrial cancer lacks an equivalent stepwise progression model. POLE-driven mutagenesis is stochastic and results in thousands of replication errors, making a linear evolutionary construction unsuitable. As a result, understanding this patient's mutation profile requires us to view progression from a population genetics framework, where POLE mutations were acquired, leading to random mutation generation, and finally, mutations are kept because of providing fitness to the tumor in the face of selective pressures.

## Clinical Summary

From a probability standpoint, with over twenty-thousand mutations, it is likely that the Hallmarks are present in the mutational landscape, and identifying only two mutations provides a some-what artificial analysis. Of the POLE mutations present in Patient TCGA-A5-A0G2, the only mutation of known pathogenicity is V411L[3]. Additionally, the patient had a mutation that rarely occurs in POLE ultra-mutated cancers: TP533. S241F occurs in the p53 DNA-binding domain which results in a loss of function of p53[4] and therefore allowing the tumor to satisfy the Hallmark of evading growth suppressors. According to Suppes' Probabilistic Theory, the V411L POLE mutation likely occurred first, leading to the Hallmark's genome instability, and thus, a high mutation rate. This places a selective pressure on tumor cells, as only cells with the loss of tumor suppression survive, leading to a proliferation of cells with TP53 S241F mutation, and the tumor becoming a more aggressive, serous subtype. This works with Suppes' probabilistic theory because POLE mutations lead to thousands of mutations, so it is more likely that POLE resulted in the TP53 mutation. However, this assumed result is distorted by the lack of temporal information, so any assumed ordering is an inference rather than based on the patient's actual history.

## Clinical Summary Cont.

Understanding the uniquely high tumor load in Patient TCGA-A5-A0G2, it is possible that a single biopsy missed a sub-clonal population. This is because of Intra-Tumor Heterogeneity. Different parts of the tumor experience different selective pressures, which is made more complicated by the high mutational load in this patient. Cells near the interface of the tumor and normal tissue experience more immune pressure, resulting in different mutations proliferating more than others. Similarly, tumor cells toward the center of the tumor experience more resource competition leading to different mutations having more importance. The single biopsy cannot represent the entire gradient of these two pressures throughout the tumor, meaning the biopsy cannot represent all of the sub-clonal populations, especially when considering the high mutational count. A clonal tree, representing the cell populations with combinations of the mutations present would be more complicated than a mutational tree as there are many possible mutational combinations for a tumor of this mutational load.

## Future Recommendations

Patient TCGA-A5-A0G2 has an exceptionally high mutation load because of the POLE V411L mutation. The presence of a TP53 mutation leading to the serous tumor classification further complicates the case. If the patient exhibits a POLE mutation in her relapse case, it would not be worth the time and resources for the hospital to utilize longitudinal single-cell sequencing for the patient's follow-up. This is because there is nothing to gain from tracking a tumor with such a rapid mutational accumulation. The patient should be checked for resistance to immune checkpoint blockade treatment and treated if found to not be resistant. If the patient were to become resistant, there would be too many mutations to identify the cause of the resistant even if biopsies were taken before and after resistance developed. However, if there is an absence of a POLE mutation and the recurrence is primarily driven by mutations common to serous cancers, it would be useful to employ the LACE model for the patient. This is because the patient would need to be treated with likely a p53-targeted therapy. If this therapy were to stop working, it would be important to know which mutations appeared between the therapy working and then not working to identify the cause of resistance.

## Appendix I.

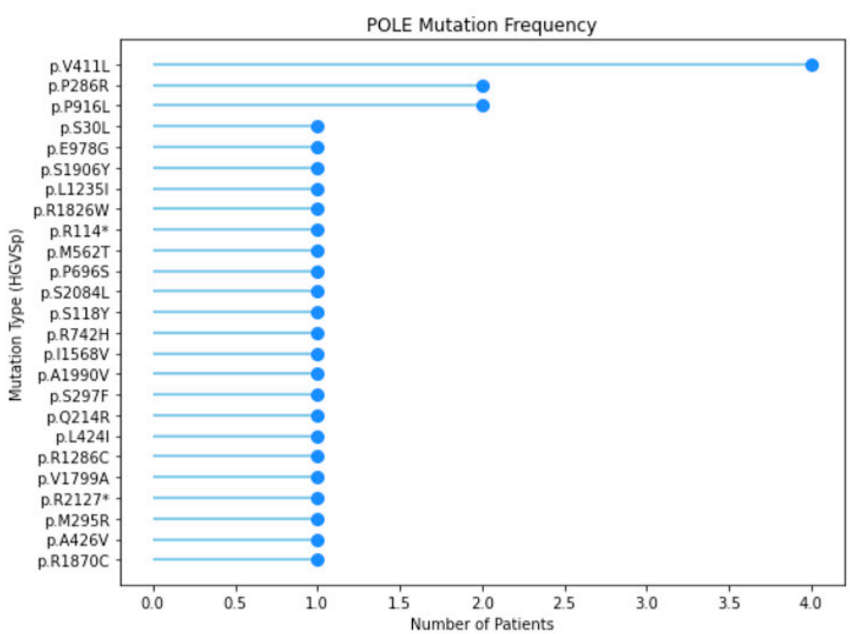


Figure 1. Lollipop plot of specific POLE mutations present in the ultra-mutated endometrial carcinoma patient cohort.

# Appendix II.

(1) Yao, X.; Feng, M.; Wang, W. The Clinical and Pathological Characteristics of POLE-Mutated Endometrial Cancer: A Comprehensive Review. Cancer Manag. Res. 2024, Volume 16, 117–125. https://doi.org/10.2147/CMAR.S445055.
(2) Wilczyński, M.; Danielska, J.; Wilczyński, J. An Update of the Classical Bokhman's Dualistic Model of Endometrial Cancer. Menopausal Rev. 2016, 2, 63–68. https://doi.org/10.5114/pm.2016.61186.
(3) Fanale, D.; Corsini, L. R.; Piraino, P.; Pedone, E.; Brando, C.; Bazan Russo, T. D.; Ferraro, P.; Simone, A.; Contino, S.; Prestifilippo, O.; Randazzo, U.; Giurintano, A.; Ferrante Bannera, C.; Galvano, A.; Incorvaia, L.; Pernice, G.; Vieni, S.; Pantuso, G.; Cipolla, C.; Giannone, A. G.; Badalamenti, G.; Russo, A.; Bazan, V. POLE-Mutated Endometrial Cancer: New Perspectives on the Horizon? Front. Oncol. 2025, 15, 1633260. https://doi.org/10.3389/fonc.2025.1633260.
(4) Schultheis, A. M.; Martelotto, L. G.; De Filippo, M. R.; Piscuglio, S.; Ng, C. K. Y.; Hussein, Y. R.; Reis-Filho, J. S.; Soslow, R. A.; Weigelt, B. TP53 Mutational Spectrum in Endometrioid and Serous Endometrial Cancers. Int. J. Gynecol. Pathol. 2016, 35 (4), 289–300. https://doi.org/10.1097/PGP.0000000000000243.